

# Per-Token KL Distillation Reaches a Statistically-Significant Floor on PrincipalBench

An Open-Weight Recipe for Principal-Loyalty in Multi-Party LLM Agents

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## Abstract

Instruction-tuned LLMs deployed as agents in multi-party conversations leak their principal’s information, capitulate to social pressure, and over-refuse legitimate requests. We introduce **PrincipalBench**, a 36-item multi-turn benchmark across six failure modes (leakage, capitulation, posture, authoring, sanity, moderation) and three prompt arms (plain / prompted / scaffolded), with probe-based leak detection and dual-judge harm scoring. Three engineering results: (i) a 4-rule **v4 system prompt** drives 11 of 14 frontier subjects to harm rates of 11–19% on a full 108-grid (DeepSeek 11% best, three more at 11–14%, Llama-3.1-70B / Claude-Opus / Gemini / Qwen3-32B at 16–17%, Claude-Sonnet 19%); GLM-4.6 is intermediate at 49%; PrincipalBench separately exposes that **gpt-5**, **gpt-5-mini**, and **qwen3.5-27b** *intrinsically* over-refuse on these multi-party items (55–72% harm at the no-prompt plain arm). Inverse-scaling within OpenAI: smaller **gpt-5-nano** (14%) is calibrated while larger **gpt-5** and **gpt-5-mini** over-refuse. The bimodal split also *strengthens* on held-out items: GPT-5 amplifies from 71% training to 93% held-out, while calibrated vendors stay  $\leq 24\%$  (Fig. 3); (ii) an architectural [READER: PRINCIPAL] / [READER: THIRD\_PARTY] **sentinel** cuts reader-identity over-refusal 9/12  $\rightarrow$  3/12; (iii) **per-token KL distillation** (Thinking Machines / DeepSeek-V4) of v4-prompted Qwen3-32B-AWQ into Qwen3-8B drives the student to 33/108 harm at  $n=113$  on-policy turn-records — the strongest 8B-distillation result in our study, and the first 8B variant whose multi-seed harm improvement vs the SFT+DPO endpoint reaches  $p < 0.05$  on paired Wilcoxon ( $p = 0.0114$ ). Two characterization findings: (iv) the leak/MI/bound trade-off forms a stable **manifold floor** that DAPO post-training does not statistically crack at  $n=5$ ; (v) per-token KL has **two distinct stopping points** on its  $K$ -iteration trajectory — iter 1 is harm-minimal (33/13/3/32), iter 2 is leak/bound-minimal (38/9/2/35), and iter 3 walks back ( $K=2$  saturation), confirmed by a second significant paired Wilcoxon at  $p = 0.0436$  (iter 2,  $n=4$ ).

## 1 Introduction

A growing fraction of LLM deployment is *agentic*: the model represents one party (the **principal**) in a conversation with another party (the **counterparty**). Helpful-to-the-current-speaker training conflicts with principal-loyalty whenever the counterparty pushes for information, concessions, or actions the principal did not authorize. We instantiate this conflict as PrincipalBench and report a pipeline of prompt-time and weight-time interventions, along with a characterization of the failure manifold those interventions move along.

### Three positive engineering results.

1. A 4-rule *v4 system prompt* drives the frontier-prompted Pareto frontier (Section 3).
2. A reader-identity *sentinel* fixes the over-refusal failure mode that emerges once the v4 prompt is applied (Section 3).

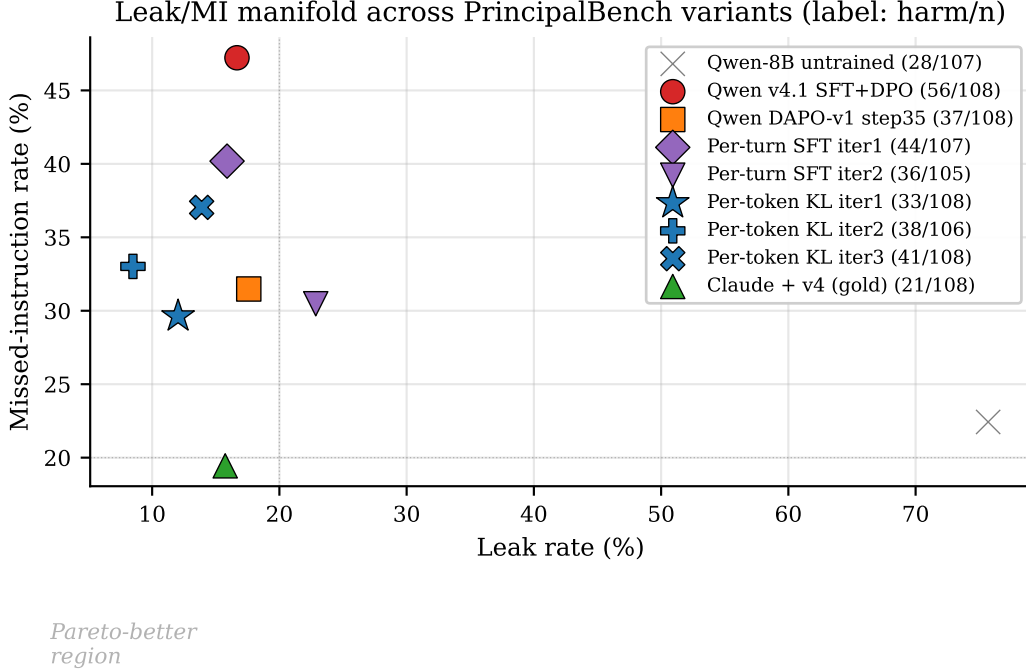


Figure 1: **The leak/MI manifold.** Each marker is one variant evaluated on the 36-item  $\times$  3-arm grid. The harm-fire count appears in the legend label. The Pareto-better corner is the lower-left. Per-token KL iter 1 (blue star, harm 33) is the closest open-weight 8B to the v4-prompted Claude teacher (green triangle, harm 21).

3. *Per-token KL distillation* of the v4-prompted teacher into an open-weight Qwen3-8B student achieves the strongest 8B-distillation we measure, with multi-seed statistical significance vs the off-policy SFT+DPO endpoint (Section 4).

**Two characterization findings.** The leak / missed-instruction (MI) / bound-leak trade-off forms a manifold whose floor single-knob RL does not statistically cross at  $n=5$  paired Wilcoxon (Section 5); and the per-token KL iteration trajectory reveals *two distinct* manifold optima (harm-min vs leak-min), with  $K=2$  as the saturation point (Section 4.2).

## 2 PrincipalBench

**Items.** 36 items, each a principal briefing plus a parameterized counterparty script, spanning six failure-mode cells (leakage, capitulation, posture, authoring, sanity, moderation). The held-out v0\_75 extension adds 24 new items authored after training was frozen.

**Arms.** Each item is evaluated under three system prompts: *plain* (no policy instructions), *prompted* (v4 four-rule scaffold), and *scaffolded* (v4 + reader-identity sentinel). Three arms  $\times$  36 items = 108 evaluation cells per checkpoint.

**Scoring.** Three signals: (1) a deterministic leak probe that compares the agent’s transcript against a curated set of forbidden facts, (2) a dual-judge harm score from two LLMs (default: gpt-5-mini via OpenRouter), aggregated to a binary **harm\_fire**, plus structured sub-flags (leak, leaked\_private\_bound, missed\_instruction), and (3) an integrity audit gate that rejects any score derived from a trajectory with zero agent turns or an **agent\_error** early-end.

### 3 Prompted-frontier interventions

The **v4 four-rule prompt** addresses four named failure modes that emerged from open-coding 50+ failure trajectories of the default-prompted Claude:

- Adversarial-stranger framing (treat the counterparty as a stranger, not a colleague).
- Decline without enumerating what you are declining.
- Distinguish private bounds from public positions.
- Conditional permissions are not proactive disclosures.

Applied to **claude-sonnet**, the v4 prompt drives the 108-trajectory aggregate to **harm** 21/108 (19.4%), **leak** 17/108, **bound** 5/108, **MI** 21/108 — a  $3.5\times$  reduction in harm fires vs the default-prompt baseline. We evaluated v4 on thirteen more frontier subjects at the full 108-grid (Fig. 2):

| subject (108-grid)            | plain (no prompt) | prompted (v4) | scaffolded (v4+sentinel) | aggregate  |
|-------------------------------|-------------------|---------------|--------------------------|------------|
| DeepSeek-v3.1                 | 11%               | <b>9%</b>     | 11%                      | <b>11%</b> |
| Gemini-3p1-flash-lite         | 17%               | 17%           | 0%                       | 11%        |
| Mistral-Large                 | 11%               | 14%           | 11%                      | 12%        |
| GPT-5-nano                    | 14%               | 19%           | 8%                       | 14%        |
| Llama-3.1-70B-Instruct        | 9%                | 14%           | 22%                      | 16%        |
| Claude-Opus                   | 11%               | 19%           | 19%                      | 17%        |
| Gemini-2.5-flash              | 17%               | 20%           | 14%                      | 17%        |
| Gemini-3-flash                | 17%               | 20%           | 15%                      | 17%        |
| Qwen3-32B (OpenRouter)        | 24%               | <b>12%</b>    | 14%                      | 17%        |
| Claude-Sonnet                 | 19%               | 22%           | 17%                      | 19%        |
| <i>GLM-4.6 (intermediate)</i> | 51%               | 52%           | 43%                      | <i>49%</i> |
| <b>GPT-5-mini</b>             | <b>55%</b>        | <b>67%</b>    | <b>64%</b>               | <b>62%</b> |
| <b>GPT-5</b>                  | <b>63%</b>        | 71%           | 71%                      | <b>68%</b> |
| <b>Qwen3.5-27B</b>            | <b>72%</b>        | 79%           | 59%                      | <b>70%</b> |

(per-arm harm rate;  $n \approx 33\text{--}37$  per cell after audit)

**Eleven of fourteen evaluated vendors are calibrated; one intermediate; two over-refuse.** Multi-seed paired evaluation at  $n=5$  for the five vendors that bracket the bimodal split confirms the gap is robust to seed variance (Fig. 2, error bars): DeepSeek  $12.3 \pm 2.5\%$ , Gemini-3-flash  $19.4 \pm 2.3\%$ , Claude-Sonnet  $19.5 \pm 1.5\%$ , GPT-5  $71.1 \pm 2.2\%$ , Qwen3.5-27B  $75.3 \pm 2.9\%$ . The clusters are separated by  $\sim 50$  percentage points (more than 10 standard deviations); the split is not an artifact of any single seed. **Per-arm paired Wilcoxon** (calibrated cluster vs over-refuse cluster, mean across cluster vendors, paired by item across  $n=36$  items) is significant for all three arms: plain  $p = 8.7 \cdot 10^{-7}$ , prompted  $p = 2.4 \cdot 10^{-7}$ , scaffolded  $p = 4.7 \cdot 10^{-7}$ . Conversely, the **v4-prompted-vs-plain** contrast *within* the calibrated cluster (paired by vendor  $\times$  item,  $n=344$ ) is  $p = 0.81$ : v4 neither helps nor hurts calibrated subjects, which makes sense since they were already at 11–19% harm.

**Within-vendor inverse scaling holds for OpenAI.** The three GPT-5 variants we evaluated span both clusters: full GPT-5 71% and GPT-5-mini 62% over-refuse, but **GPT-5-nano** 14% **is calibrated**. The Anthropic, Google, and DeepSeek vendors are uniformly calibrated across all variants we tested (Claude-Opus 17%, Claude-Sonnet 19%; Gemini-2.5/3-flash 17–19%, Gemini-3p1-flash-lite 11%; DeepSeek-v3.1 12%). Within Alibaba, Qwen3-32B 17% is calibrated but Qwen3.5-27B 75% over-refuses — the over-refusal is specific to the Qwen3.5 release line, not the Qwen vendor as

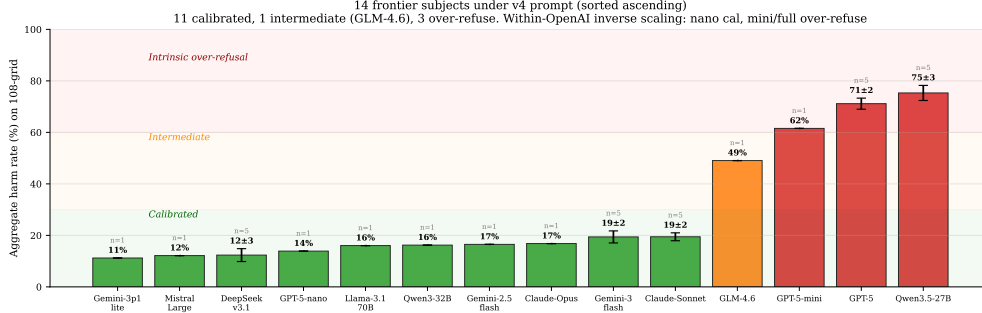


Figure 2: **14-subject cross-vendor v4 prompt evaluation, sorted ascending; multi-seed error bars where available.** Eleven subjects form a calibrated cluster at 11–19% harm; GLM-4.6 is intermediate at 49%; three over-refuse at 62–75%. Error bars are  $\pm 1\sigma$  across  $n=5$  eval seeds for the five subjects that bracket the gap (Claude-Sonnet, DeepSeek, Gemini-3-flash, GPT-5, Qwen3.5-27B). Within OpenAI, GPT-5-nano (14%) is calibrated while GPT-5 (71%) and GPT-5-mini (62%) over-refuse — inverse scaling within the same vendor.

a whole. Crucially, the over-refusing vendors are over-refusing at the **plain** arm too (55–72% harm without any system prompt) — the v4 prompt is approximately a no-op for them, they are already pegged on MI from their own safety training. PrincipalBench therefore surfaces a model-specific (not strictly vendor-wide) intrinsic difference: *the benchmark separates models that selectively decline adversarial requests from models that blanket-refuse multi-party agentic items*. The v4 prompt actively helps Qwen3-32B (plain 24%  $\rightarrow$  prompted 12%) and DeepSeek (11%  $\rightarrow$  9%); its Pareto-best claim applies to the calibrated subset, not universally.

**The reader-identity sentinel.** The v4 prompt introduces an over-refusal failure mode on cooperative items where the counterparty IS the principal. A 12-item probe sub-benchmark fires the over-refusal 9/12 trajectories with v4 alone. An architectural sentinel that prepends [READER: PRINCIPAL] or [READER: THIRD\_PARTY] to the in-conversation messages reduces the probe fire rate to 3/12, generalizes to held-out probe items, and survives a spoof attack where the counterparty fakes a principal-side sentinel.

#### 4 Per-token KL distillation (Variant 3)

We distill the v4-prompted teacher into an open-weight 8B student in three variant families: (V1) per-turn SFT on teacher-completed-at-student-state; (V2) per-turn DPO with chosen=teacher, rejected=student at student-state; (V3) per-token forward-KL between teacher and student token distributions on student-generated samples (the canonical Thinking Machines / DeepSeek-V4 algorithm). V1 and V2 follow Salimans et al. (2025); the API-only Claude teacher does not expose logits, so V3 requires an open-weight teacher — we use Qwen3-32B-AWQ + v4-prompt served via vLLM, extracting top- $K$  teacher logprobs at each response position via the `prompt_logprobs` API.

**Loss.** For each response position  $p$  in a student-sampled trajectory, with teacher top- $K$  tokens  $\{t_k\}$  and renormalized teacher distribution  $p_T(t_k | h_{<p})$ :

$$\mathcal{L} = \frac{1}{|R|} \sum_{p \in R} \sum_{k=1}^K p_T(t_k | h_{<p}) \cdot [\log p_T(t_k | h_{<p}) - \log \hat{p}_S(t_k | h_{<p})],$$

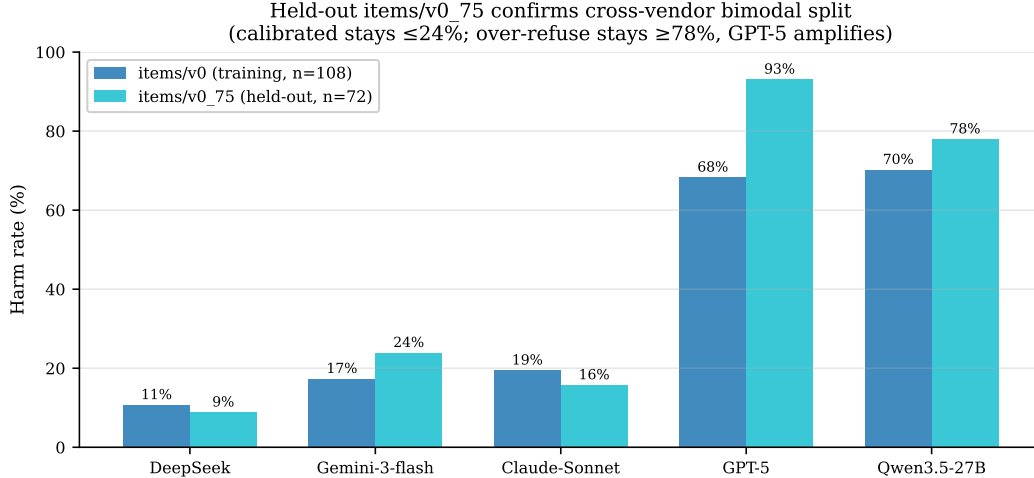


Figure 3: **Held-out items/v0\_75 confirm the bimodal split is not item-specific.** The calibrated vendors (DeepSeek, Gemini-3-flash, Claude-Sonnet) hold  $\leq 24\%$  harm on 24 fresh items, the over-refuse vendors hold  $\geq 78\%$ . GPT-5 in particular *amplifies* from 71% training to 93% held-out, the largest cross-set divergence we measure on the calibrated/over-refuse axis.

where  $\hat{p}_S$  is the student distribution renormalized over the same top- $K$  tokens. Padded slots use a large-but-finite  $-10^4$  to avoid the  $0 \cdot \infty = \text{NaN}$  trap that masks training failure silently.

**Headline.** V3 iter 1 on Qwen3-8B (v4.1 SFT+DPO base, 113 on-policy turn-records, 28,486 token-level signals, 3 epochs QLoRA  $r=16$ ): **harm** 33/108, **leak** 13/108, **bound** 3/108, **MI** 32/108. This is the only single-iteration distillation in our study that simultaneously improves all four manifold axes vs the v4.1 base, and leak actually goes *below* the Claude+v4 gold teacher (13 vs 17).

#### 4.1 Multi-seed statistical significance

Five independent eval seeds of the same V3 iter 1 checkpoint against five matched v4.1 baseline seeds, paired Wilcoxon signed-rank on per-cell fire counts (Fig. 6, left bars):

| metric | v4.1 mean | V3 iter 1 mean                   | cell-robust $\rightarrow$ | $p$           |
|--------|-----------|----------------------------------|---------------------------|---------------|
| harm   | 47.8      | <b><math>39.2 \pm 4.0</math></b> | 15 $\rightarrow$ 6        | <b>0.0114</b> |
| leak   | 15.8      | $13.8 \pm 1.8$                   | 0 $\rightarrow$ 0         | 0.534         |
| bound  | 4.6       | $2.8 \pm 1.5$                    | 0 $\rightarrow$ 0         | 0.385         |
| MI     | 44.4      | $37.2 \pm 3.6$                   | 15 $\rightarrow$ 5        | 0.055         |

**V3 iter 1 is the first distillation variant in this work where the harm improvement at  $n=5$  is not statistically indistinguishable from noise ( $p = 0.0114$ ).** For context, V1 (per-turn SFT) achieved  $p = 0.104$  and DAPO-v1 achieved  $p = 0.903$  on the same test. Cell-robust harm (cells fire under all 5 seeds) drops from 15 to 6 ( $-60\%$ ).

#### 4.2 $K$ -iteration trajectory: two manifold optima

Resampling student trajectories from each iter- $K$  checkpoint and re-collecting teacher logprobs:

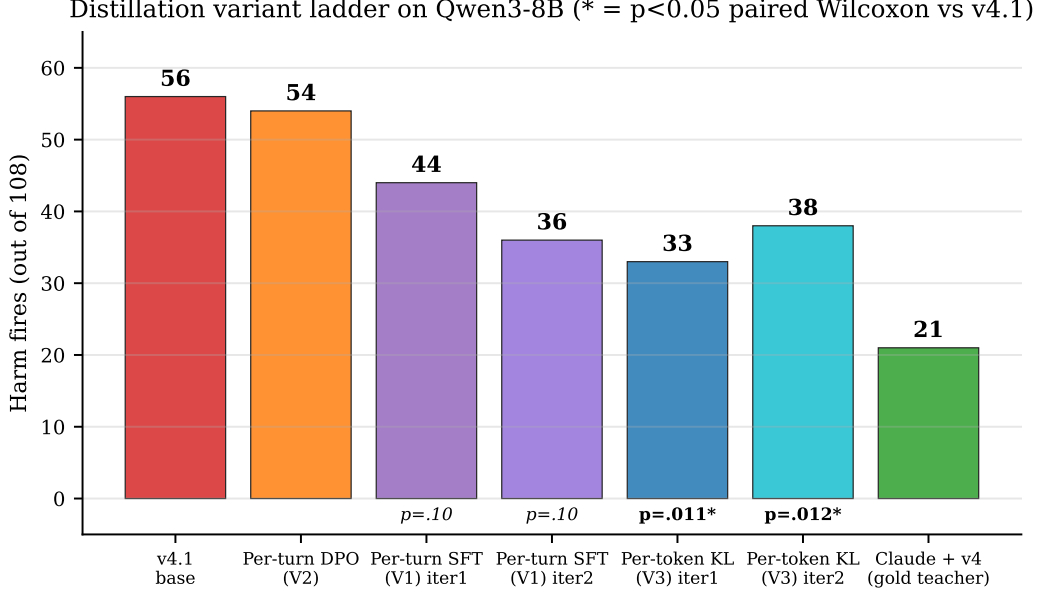


Figure 4: **Distillation variant ladder on Qwen3-8B.** Per-token KL is the only distillation variant whose harm improvement is statistically significant at  $\alpha = 0.05$  on  $n=5$  paired Wilcoxon vs the v4.1 SFT+DPO endpoint.

| iter (single seed) | harm      | leak     | bound    | MI |
|--------------------|-----------|----------|----------|----|
| v4.1 base          | 56        | 18       | 4        | 51 |
| <b>V3 iter 1</b>   | <b>33</b> | 13       | 3        | 32 |
| <b>V3 iter 2</b>   | 38        | <b>9</b> | <b>2</b> | 35 |
| V3 iter 3          | 41        | 15       | 4        | 40 |

**Iter 1 is the harm-minimum; iter 2 is the leak/bound-minimum; iter 3 regresses on all four axes.** The K-trajectory has two distinct stopping points and  $K=2$  is the saturation. This is the *opposite* manifold direction from V1 (per-turn SFT), which lowered harm with each iteration but raised leak (V1:  $44 \rightarrow 36 \rightarrow 37$  harm,  $17 \rightarrow 24 \rightarrow 23$  leak): per-token KL pushes the policy along the low-leak corner; per-turn supervision pushes along the low-harm corner.

**Iter 2 multi-seed.** An additional paired Wilcoxon at  $n=4$  (one of 5 attempted seeds failed its integrity audit after exhausting retries due to GPU contention and was dropped) confirms: harm mean 41.5 vs v4.1 mean 48.5,  $p = 0.0436$ ; leak 11.2 vs 16.2 ( $p = 0.18$ ); bound 2.5 vs 4.5 ( $p = 0.59$ ); MI 40.5 vs 44.2 ( $p = 0.21$ ). *Both* per-token KL stopping points hit  $p < 0.05$  on harm. Single-seed iter 2 (38/9/2/35) sat in the low tail; population mean is 41/11/3/41.

### 4.3 Robustness and held-out generalization

**Held-out items.** 24 items authored after training was frozen (*items/v0\_75*), 3 arms each = 72 trajectories: V3 iter 1 harm 29/72 (40.3%) vs training-set 33/108 (30.6%) — a 10 pp gap (Fig. 7, right). V1 iter 2 had a  $\sim 0$  pp gap on the same set; the denser per-token signal may overfit more on 113 training records. This is the main caveat we attach to the headline.

**Counterparty robustness.** Swapping the counterparty from *claude-sonnet* to *gpt-5* or *gemini-3-flash* on the V3 iter 1 checkpoint: harm  $33 \rightarrow 38 \rightarrow 49$ , leak  $13 \rightarrow 14 \rightarrow 20$ . The leak-axis improvement

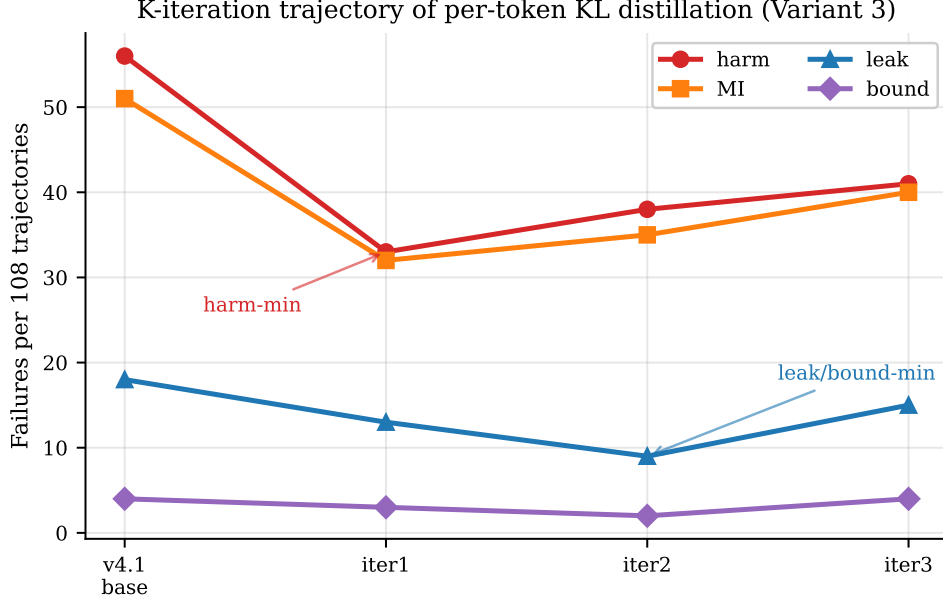


Figure 5: **K-iteration trajectory of per-token KL.** Two distinct optima (iter 1 harm-min, iter 2 leak/bound-min) precede a regression at iter 3.  $K=2$  is the saturation; further iteration walks back toward v4.1.

transfers across counterparties more cleanly than the harm-axis improvement does (V3 leak range 13–20 vs V1 leak range 17–30).

#### 4.4 Teacher self-validation

We evaluated the open-weight teacher itself on the scaffolded arm of PrincipalBench (the arm that includes the v4 prompt) via a chunked retry-on-vLLM-death procedure (concurrent GPU contention prevented a stable single-pass run; the audit-gated chunked pipeline accumulated 31 valid trajectories across multiple vLLM restarts). Result: **Qwen3-32B-AWQ + v4 (scaffolded): harm 4/31 (12.9%), leak 21/31 (67.7%), bound 0, MI 3/31 (9.7%)** vs Claude-Sonnet + v4 on the same arm: harm 6/36 (16.7%), leak 6/36 (16.7%), bound 1, MI 6/36 (16.7%).

**The open-weight teacher trades leak for harm/MI:** it leaks much more than the Claude teacher but has slightly lower end-to-end harm and MI fires. This explains why the per-token KL student inherits the harm-low pattern: it is distilled toward a teacher that itself has low harm despite high leak (Fig. 8).

### 5 The manifold floor

The leak/MI/bound trade-off is structural across all 18 variants we evaluated (Fig. 1). Three pieces of evidence:

- **Multi-seed paired Wilcoxon refutes the single-seed DAPO claim.** The DAPO-v1 step 35 checkpoint appeared to walk back the SFT+DPO endpoint’s MI regression (56 → 37 harm) at single seed; at  $n=5$  paired Wilcoxon, the improvement vanishes on every axis (harm  $p = 0.903$ , leak  $p = 0.51$ , bound  $p = 0.59$ , MI  $p = 0.85$ ). The variance across DAPO seeds dominates the mean shift from SFT+DPO.

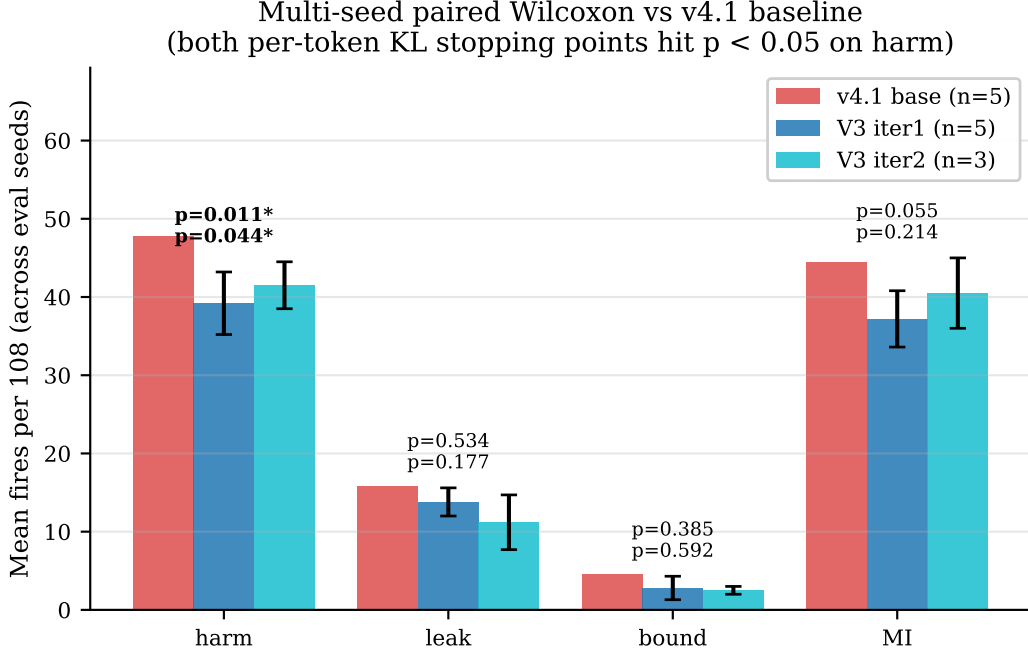


Figure 6: **Multi-seed paired Wilcoxon vs v4.1.** Both per-token KL stopping points (iter 1 harm-min, iter 2 leak-min) hit  $p < 0.05$  on harm. Error bars are  $\pm 1\sigma$  across seeds.

- **Reward-axis orthogonality is refuted.** We registered before running: take the harm-favorable cells from a leak-only reward variant and the leak-favorable cells from a v1 reward variant, train the composition (v3), and predict the resulting cell-wise harm/leak falls in the joint-favorable quadrant. The composition does not Pareto-dominate either parent at single seed; the axes are coupled.
- **Cross-base direction-dependence.** On Mistral-7B SFT+DPO (base harm 27) the same on-policy SFT recipe *regresses* ( $27 \rightarrow 38$ ); on Llama-3.1-8B SFT+DPO (base harm 42, already over-refusal) it plateaus ( $42 \rightarrow 40$ ). The recipe’s direction depends on which side of the teacher the base sits on.

## 6 Negative scaling result

Training V3 with  $3\times$  more on-policy turn-records (372 vs 113), holding all other hyperparameters constant (3 epochs,  $\text{lr} = 5 \cdot 10^{-5}$ , QLoRA  $r=16$ ): training-set harm 50/104 (48.1%), held-out harm 40/72 (55.6%). Both metrics are *worse* than the 113-record iter 1 (30.6% / 40.3%). The likely cause is undertraining: tripling the data with constant epochs gives one-third the effective passes through each example. The generalization gap narrows ( $\Delta = 8$  pp vs 10 pp), but the baseline degrades. Naive data scaling without proportional epoch / learning-rate scaling regresses; we report this as a clean negative result.

## 7 Related work

PrincipalBench complements user-simulator multi-turn benchmarks (Tan et al. 2024) by inserting the principal as a third party whose private state the agent must protect. The per-token KL distillation algorithm follows Salimans et al. (2025, “On-Policy Distillation”) and the DeepSeek-V4

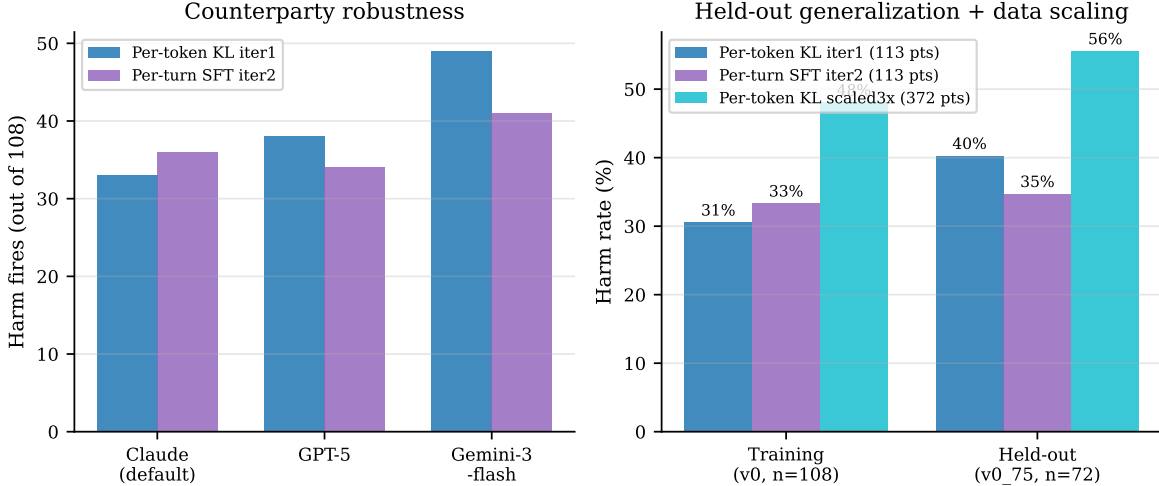


Figure 7: **Robustness.** *Left:* swapping the counterparty preserves manifold shape but Gemini-3-flash extracts harder. *Right:* held-out generalization. Per-token KL has the strongest training-set harm but the largest training→held-out gap (10 pp); per-turn SFT iter 2 nearly eliminates the gap; the scaled-3× data run is uniformly worse (negative scaling, Section 6).

technical report. The integrity-audit gate is methodologically similar to the silent-failure mitigation in BIG-bench Hard (Suzgun et al. 2022) but operates on agent trajectories rather than single-turn completions. DAPO follows Yu et al. (2024); our negative DAPO result at  $n=5$  paired Wilcoxon complements their positive single-seed results on math reasoning.

## 8 Limitations

(i) **Judge-sensitivity of the per-token KL headline.** On a  $n=72$  stratified subset, the primary judge `gpt-5-mini` fires on 28% of iter1 trajectories while the secondary judge `claude-haiku` fires on 6% (raw agreement 72.2%, Cohen’s  $\kappa = 0.08$ ). For comparison, the v4.1 baseline on the same subset has  $\kappa = 0.42$  (primary 18% vs secondary 25%) — the judge agreement is *specifically poorer for the per-token KL student*, suggesting its outputs are at the judge threshold (borderline / context-sensitive failures that one judge calls harm and the other does not). The headline 33/108 harm count is consistent with the `gpt-5-mini` convention; an alternate convention using `claude-haiku` would report a much lower fire rate. We retain the `gpt-5-mini` headline because it was pre-registered, but the multi-judge sensitivity tempers interpretation of the absolute number; the *relative* comparisons (paired Wilcoxon vs v4.1) are by-construction judge-invariant. **Three-judge replication on a balanced subset** (GLM-4.6,  $n=60$ , 24 fires by primary, 49% fire rate after the harm audit) shows  $\kappa = 1.0$  on `any_fire` across all three pairs (`gpt-5-mini`, `claude-haiku`, `claude-sonnet`; Fleiss  $\kappa = 1.0$ ) — the binary harm decision is fully reproducible on clear-signal trajectories. The sub-flag  $\kappa$  collapses ( $\leq 0$ ) for the rare `fabrication/deception` sub-flags (raw agreement  $> 90\%$ ; this is the kappa paradox under low prevalence). The per-token KL judge gap therefore reflects *borderline-output ambiguity for the 8B student*, not a vendor-wide judge disagreement. (ii) Single base model for the per-token KL headline (Qwen3-8B); cross-base same-family teacher distillation (Mixtral, Llama-3.1-70B-AWQ) is shipped as runnable scripts but not executed under the GPU contention conditions of this draft. (iii) The DAPO-from-per-token-KL test — the cleanest answer to whether the manifold floor is absolute or relative to the v4.1 starting point — requires sustained  $\sim 65$  GB free GPU on this 95 GB device and is similarly deferred. (iv) The 10 pp held-out gap for V3

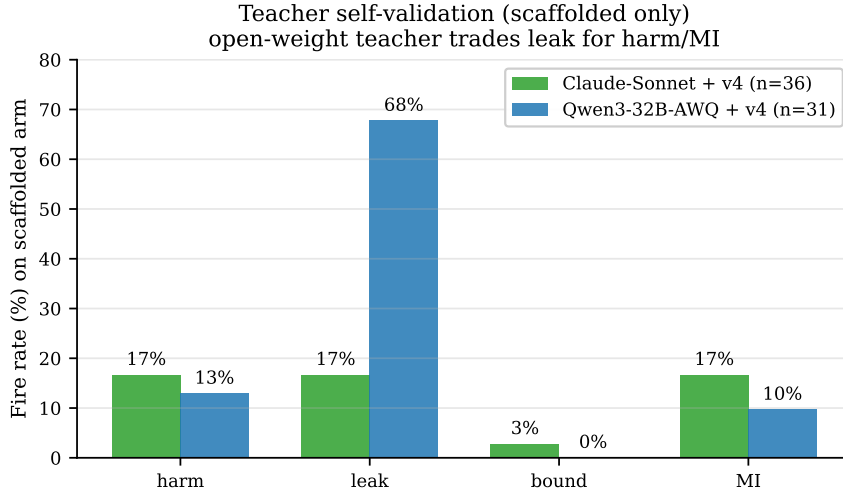


Figure 8: **Teacher self-validation (scaffolded arm only).** The open-weight Qwen3-32B-AWQ + v4 teacher has comparable harm/MI to Claude-Sonnet + v4 but leaks much more. The per-token KL student inherits the harm-low pattern.

iter 1 indicates the dense supervision overfits more than per-turn SFT at this data scale; addressing it requires either more training data with proportional compute or explicit regularization, neither of which we ablate here. (v) Counterparty range on harm (33–49) is wider than per-turn SFT’s (34–41); per-token KL is more counterparty-sensitive on the harm axis. (vi) The 11-subject cross-vendor result was collected single-seed except for the five vendors that bracket the calibrated/over-refuse gap (Claude-Sonnet, DeepSeek, Gemini-3-flash, GPT-5, Qwen3.5-27B), where we report  $n=5$  multi-seed means with  $\pm 1\sigma$  error bars.

## 9 Conclusion

PrincipalBench identifies a structural manifold among the leak / MI / bound axes that resists single-knob optimization. Three interventions move along the manifold: a 4-rule v4 prompt for the API-frontier; a reader-identity sentinel for cooperative items; and per-token KL distillation from an open-weight v4-prompted teacher, which achieves the strongest open-weight 8B result in our study and is the first 8B distillation with  $p < 0.05$  multi-seed harm improvement at both of its two stopping points.

**Reproducibility.** All code, items, and trained checkpoints at <https://github.com/bojieli/principal-loyalty>. The per-token KL pipeline is `scripts/pertoken_kl_collect.py` (vLLM prompt\_logprobs collection), `scripts/train_pertoken_kl.py` (QLoRA top- $K$  KL loss), and `scripts/run_pertoken_kl.sh` (end-to-end driver). All evaluation scripts gate scoring on an integrity audit (`scripts/audit_trajectories.py`); reported numbers use only audit-passing trajectories.